

# CHUHAO HUANG 黄楚昊

## CONTACT INFORMATION

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Département de Mathématiques et Applications  
École Normale Supérieure  
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## RESEARCH INTERESTS

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My research interests lie at the interface between algebraic topology and algebraic geometry, with a particular focus on applying techniques from cobordisme theory to the study of algebraic cycles.

**Keywords:** Algebraic cycles, Complex cobordism, Algebraic cobordism, Hodge theory, etc.

## CURRENT POSITION

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**École Normale Supérieure**

*Sep 2024 - expected July 2027*

Currently, I'm pursuing a Ph.D. in mathematics at ENS under the supervision of [Olivier Benoist](#).

*I will be on the job market in Fall 2026.*

## EDUCATION

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**École Normale Supérieure**

*Sep 2023 - July 2024*

*Préthèse* in Mathematics, under the supervision of [Olivier Benoist](#).

**École Normale Supérieure & Sorbonne Université**

*Sep 2021 - July 2023*

*Normalien* via *Sélection Internationale*

M.S. in Mathematics

Mentions pour M1 et M2: Très Bien.

**Peking University**

*Sep 2018 - July 2022*

B.S. in Mathematics, supervised by [Bohan Fang](#).

## RESEARCH

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**Smoothing low-dimensional cycles in algebraic cobordism**

*Year 2026*

Preprint, arXiv: [2605.04347](#).

## THESIS AND NOTES

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**Quillen's Theorem in Complex Cobordism and its Applications**

*Year 2023*

*Mentored by Olivier Benoist*

M2 thesis at École Normale Supérieure and Sorbonne Université: [Here](#).

**Arbres et Amalgames**

*Year 2022*

*Mentored by Gaëtan Chenevier, jointed work with Yuxiao Xie*

First-year thesis (*Mémoire de première année*) at École Normale Supérieure: [Here](#).

**Topics in Model Categories**

*Year 2022*

*Mentored by Bohan Fang*

Undergraduate thesis at Peking University: [Here](#).

## Mirror Symmetry of Elliptic Curves

Year 2020

*Mentored by Owen Barrett*

Undergraduate research paper for the REU program of Chicago University: [Here](#).

A longer but more self-contained version can be found [Here](#).

## TEACHING EXPERIENCE

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**Teaching Assistant / Chargé de TD, Algebraic Topology (Niveau M1)**     *ENS, Spring 2026*

Led tutorial sessions (travaux dirigés) and prepared [exercise solutions](#).

**Undergraduate Research Supervision**     *ENS, Spring 2025*

Advised two undergraduate students on their first-year thesis (*Mémoire de première année*) in Chern–Weil theory.

## HONOR

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Freshman Scholarship, Peking University

*Year 2018*

Gold Medal, 32<sup>nd</sup> Chinese Mathematics Olympiad (CMO)

*Year 2017*

Selected for Chinese national training team.

## LANGUAGE

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Mandarin (*native*), English (*fluent*), French (*intermediate*)

## SKILL

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Python, C, C++, LaTeX